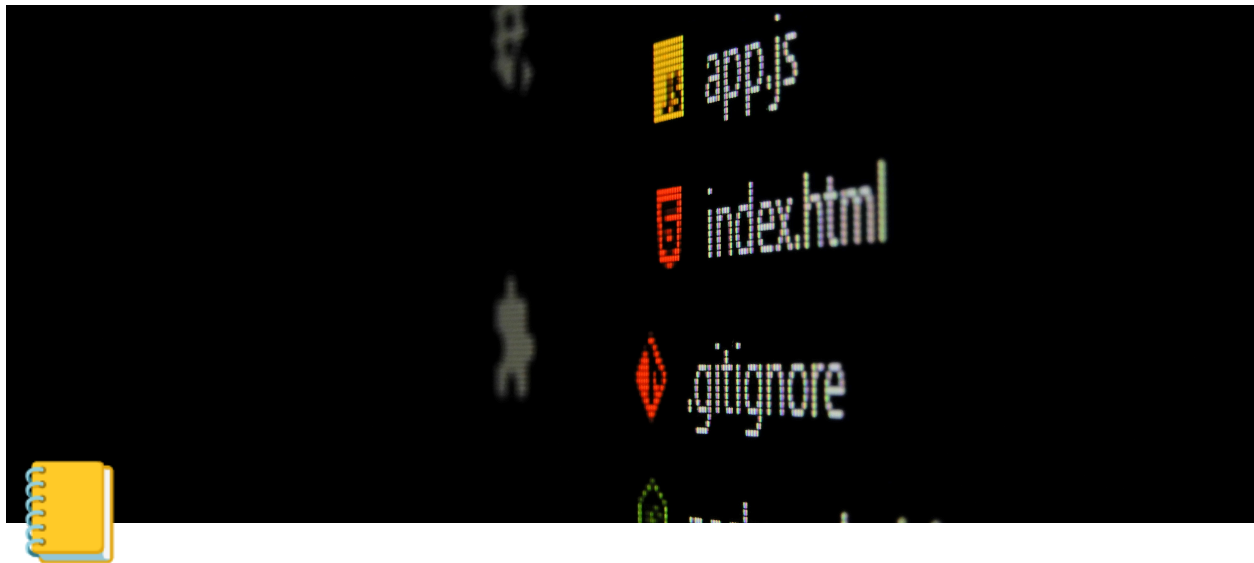




HTML Is Not What You Think

1. What Most People think about HTML(Misconceptions)

There are so many misconceptions that beginners think that

- HTML is programming language
- HTML create design
- HTML makes websites
- HTML is too basic and it is not worth it

2. What HTML Actually IS

HTML stands for Hypertext Markup Language . It is most basic building blocks of the Web. It defines the structure and meaning of the web content.

“Hypertext” - it refers to the link that connect pages to one another within single websites.

Purpose Of a Markup Language

- Defining Structure
- Annotation - They allow users to annotate (to make notes) with instructions on how it should be displayed on the web.
- SEO - Provides facility for search engine optimizations

Meaning of markup in HTML

"Markup" - refers to the tags used to define elements within a document. These tags show that how the content should be displayed . It includes various tags such as `<h1>` to `<h6>` (for Headings), `<p>` (for paragraphs) , `<img>` (for images) and more.

Markup in HTML is enclosed in angular brackets (`<` `>`) .

Browser reads HTML like a document tree , and we will see what happens behind the scenes when browser loads a website .

Evolution of HTML

3. HTML Is Like a Skeleton



Real Life Analogy

5. Semantic HTML

There are so many ways to create a website , and are infinite ways to mark up your content . But some ways are better than others and they are **Semantics**

Semantics means " which is related to meaning " .Means writing our HTML code that actually depicts its meaning own its own .

6. How Browsers Understand HTML

A web browser interprets HTML by parsing it to construct the Document Object Model (DOM) of a webpage

Parsing means : Breaking something into smaller parts so a computer (or a person) can understand and process it .

In simple words **Parsing** means "Reading and understanding the code."

So when a web browser receives HTML from a server, it doesn't just display the raw code. Instead, it interprets the HTML, which means it reads and understands the structure and content of the webpage as described by the HTML.

The process begins with the browser parsing the HTML, which involves breaking down the HTML document into individual elements. These elements include tags such as `<head>`, `<body>`, `<p>`, `<div>`, and many others, as well as the content within these tags. The browser reads the HTML from top to bottom, and as it encounters each element and creates a Document Object Model (DOM).



The DOM is a tree like representation of the webpage , with each node representing an HTML element. It helps browser to understand the structure of the webpage and how different elements are related to each other.

For Example :

```
<body>
    <div>
        <p>This is paragraph inside a div</p>
    </div>
```

This would result in a `'p'` node as a child of a `'div'` node in the DOM. More on this we will study in the JavaScript.

Now Once the DOM is constructed , the browser can render the webpage. This involves taking the structure and the content as described by the DOM and displaying it on the screen in the way specified by the HTML and any associated CSS.



Summary : A web browser interprets HTML by parsing it to construct the DOM which it then uses to render and interact with the webpage .

7. HTML and SEO

Up to this point as many of us now know that HTML is used to structure the content on a webpage. But besides that it also helps search engines (crawlers) to better understand your website and helps to rank them.

1. Heading Hierarchy

HTML headings help search engines understand the structure of a webpage.

```
<h1>Main Title</h1>  
<h2>Sub Topic</h2>  
<h3>Small Section</h3>
```

Search engines use headings to identify the

- Main topics
- Sub-topics
- Important Keyword



Important Note

`<h1>` heading should usually be used once for the main topic
Proper hierarchy improves readability and SEO

2. Meta Tags

Meta tags provide the information about the webpage.

They are written inside the `<head>` tag.

```
<meta name="description" content="HTML Documentation">
```

Meta tags help search engines to understand about

- about the website
- author

- character encoding

When the webpage appears on Google , the small description shown below the title often comes from meta tags.



Good meta description can improves click rates.

3. Semantic Structure

Semantic HTML means using tags according to their meaning.

Example

```
<header>
<nav>
<main>
<section>
<article>
<footer>
```

Instead of using many `<div>` tags everywhere, semantic tags clearly explain the purpose of content.

It helps SEO and engines can better understand that this is

- navigation areas
- main content
- articles
- footers
- sections

4. Alt Attribute

The `alt` attribute is used inside image tags.

```

```



Importance of Alt Attribute

- The alt text is an important attribute and good developers didn't skip because it gives us the information about the image (if in case our image does not load).
- Search engines cannot fully "see" images like humans. So they read the alt text to understand the image content.
- Visually impaired users using screen readers

8. HTML and Accessibility

HTML Is Not About Appearance